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National Examination Management System (NPTEL)

System Understanding and Database Design Documentation

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Chapter 1

System Understanding

1.1 Introduction

The National Examination Management System (NPTEL) is a centralized, database-driven information system designed to manage the complete lifecycle of the National Programme on Technology Enhanced Learning (NPTEL) – a joint initiative of the Indian Institutes of Technology (IITs) and the Indian Institute of Science (IISc), funded by the Ministry of Education, Government of India.

NPTEL delivers online certification courses (MOOCs) in engineering, science, management, and humanities disciplines through the SWAYAM platform. Learners may audit courses for free or opt for a proctored certification exam conducted at designated Test Centres. The system must handle millions of learners, thousands of courses per semester, and nationwide proctored examinations conducted twice a year (January–April and July–October semesters).

NPTEL is designed using a structured relational database model to ensure ACID properties, minimize redundancy, and maintain consistency across all interconnected modules – from course enrollment to certificate issuance.

1.2 Objectives of the System

The primary objectives of the National Examination Management System are:

- To manage large-scale learner registration, authentication, and course enrollment securely.
- To support course creation, discipline mapping, and instructor assignment by NPTEL administrators.
- To process online weekly assignment submissions and auto-compute scores throughout the course duration.
- To handle proctored examination registration and secure online fee payment.
- To allocate learners to NPTEL-approved Test Centres based on city preference and capacity.
- To ensure secure and unbiased evaluation of proctored answer sheets.
- To compute final scores (25% assignments + 75% proctored exam) and generate NPTEL certificates.
- To issue categorized certificates: Topper, Elite + Gold, Elite, and Pass, based on score thresholds.
- To facilitate systematic grievance handling post result declaration.
- To maintain high data integrity, transaction consistency, and concurrency control for a

national-scale system.

1.3 System Workflow

The operational workflow of NPTEL is divided into the following structured phases:

1.3.1 Pre-Course Phase

- Creation of semester records (Jan–Apr / Jul–Oct) by NPTEL administrators.
- Discipline and course creation with instructor assignment from IIT/IISc faculty.
- Registration of Test Centres along with city and capacity specifications.
- Activation of learner registration and course enrollment portal on SWAYAM.

1.3.2 Enrollment and Assignment Phase

- Learner registration and profile creation on the SWAYAM/NPTEL portal.
- Free enrollment into one or more NPTEL courses.
- Weekly assignment release by the system (typically 4–12 weeks per course).
- Online submission and automated grading of weekly assignments; best-of-N policy applied.

1.3.3 Proctored Exam Registration Phase

- Learners who wish to earn a certificate register for the proctored examination.
- Online payment of proctored exam fee (approximately INR 1,000 per course).
- Selection of preferred city/Test Centre; allocation based on seat availability.
- Generation of Hall Ticket (admit card) after successful payment verification.
- Assignment of halls, seat numbers, and invigilators at Test Centres.

1.3.4 Proctored Examination Phase

- Conduct of proctored examination at assigned Test Centres (typically conducted by TCS iON).
- Collection and secure dispatch of OMR/digital answer sheets for evaluation.

1.3.5 Evaluation and Result Phase

- Evaluation of proctored answer sheets by assigned evaluators.
- Computation of final score: Assignment Score (25%) + Proctored Exam Score (75%).
- Determination of certificate category: Topper, Elite + Gold, Elite, Pass, or No Certificate.
- Result declaration and certificate generation on the NPTEL portal.

1.3.6 Grievance Handling Phase

- Submission of grievances by learners after result declaration within the stipulated window.
- Review and resolution of grievances by the NPTEL academic committee.

1.4 Major Functional Modules

The system is logically divided into the following modules:

Learner Management Module

Handles learner registration, authentication, profile management, and academic background verification.

Course and Discipline Module

Manages creation of disciplines, courses, weekly assignment schedules, and instructor assignments.

Enrollment Module

Processes free course enrollments and tracks assignment submission history for each learner-course pair.

Proctored Exam Registration Module

Manages learner registration for proctored examinations and maintains registration status.

Payment Management Module

Handles secure online transactions for proctored exam fees and payment verification.

Test Centre and Allocation Module

Manages Test Centre registration, capacity tracking, hall assignment, and seat allocation for proctored exams.

Evaluation and Result Module

Assigns evaluators, records proctored exam marks, aggregates with assignment scores, and generates final results with certificate categories.

Certificate Module

Issues digitally signed certificates based on final score and category thresholds defined by NPTEL.

Grievance Management Module

Maintains records of learner complaints and tracks resolution status through the academic committee.

1.5 System Constraints

1.5.1 Integrity Constraints

- A learner can enroll in a specific course only once per semester.
- Proctored exam registration is permitted only for learners who are actively enrolled in the course.
- Hall ticket generation is allowed only after successful payment verification.
- Test Centre capacity must not exceed its defined limit during allocation.
- Each answer sheet must be evaluated by an authorized NPTEL evaluator.

- Final score is computed only after both assignment scores and proctored exam marks are recorded.

1.5.2 Business Rules

- Assignment score is the average of best N out of all weekly submissions (N defined per course).
- Proctored exam registration is optional; unenrolled learners cannot sit the proctored exam.
- Certificate category thresholds: Topper (top 2%, $\geq 90\%$), Elite + Gold ($\geq 90\%$), Elite ($\geq 75\%$), Pass ($\geq 40\%$).
- Grievances can be raised only after result publication and within the defined grievance window.
- Hall allocation is performed only for verified (paid) exam registrations.

1.5.3 Security Constraints

- Learner credentials must be stored using secure hashing mechanisms (bcrypt/SHA-256).
- Only authorized NPTEL administrators can create courses and assign evaluators.
- Learners are restricted from modifying evaluation results or assignment scores post-submission.
- Payment gateway transactions must be encrypted using TLS/SSL.

1.6 Assumptions and Scalability Considerations

For database design and modeling purposes, the following assumptions are made:

- The system supports millions of concurrent learner records across all active semesters.
- Hundreds of NPTEL-approved Test Centres are distributed geographically across India.
- Thousands of evaluators from IITs and IISc participate in answer sheet assessment.
- Each course has between 4 and 12 weekly assignments, all graded automatically.
- High concurrency is expected during peak enrollment, assignment submission, and proctored exam registration periods.
- Multiple disciplines (CSE, ECE, ME, CE, Humanities, Management, etc.) exist, each containing many courses.

The database schema must therefore support optimized indexing, a normalized structure, and efficient query execution plans.

1.7 Transaction Requirements

To ensure reliability and consistency, the system enforces ACID properties for critical operations:

- Payment processing and proctored exam registration approval must occur atomically.
- Assignment score computation and update must reflect only validated submissions.

- Final score computation must reflect accurately evaluated proctored exam marks and consolidated assignment scores.
- In case of transaction failure, rollback mechanisms must preserve data integrity.
- Concurrent enrollment submissions and payment transactions must not cause data inconsistency.

This comprehensive understanding of the system forms the foundation for identifying entities, attributes, relationships, functional dependencies, and normalization in subsequent design phases.

Chapter 2

Identification of Entities, Attributes and Constraints

Based on the detailed system analysis, the following core entities are identified for the National Examination Management System (NPTEL).

Each entity is defined with appropriate domain constraints, integrity constraints, and referential constraints to ensure data consistency, correctness, and normalization.

2.1 LEARNER

Represents individuals who register on the SWAYAM/NPTEL portal to take online courses.

Attribute	Type	Constraints
learner_id	NUMBER	PRIMARY KEY
full_name	VARCHAR2(100)	NOT NULL
dob	DATE	NOT NULL
gender	VARCHAR2(10)	NOT NULL, CHECK (gender IN ('Male','Female','Other'))
email	VARCHAR2(150)	NOT NULL, UNIQUE
phone	VARCHAR2(15)	NOT NULL, UNIQUE
aadhaar_no	VARCHAR2(12)	NOT NULL, UNIQUE
education_level	VARCHAR2(50)	NOT NULL, CHECK (IN 'School','UG','PG','PhD','Other')
institution_name	VARCHAR2(200)	NOT NULL
state_of_residence	VARCHAR2(100)	NOT NULL
password_hash	VARCHAR2(255)	NOT NULL
registration_date	DATE	DEFAULT SYSDATE, NOT NULL

Key Constraints:

- Email, phone, and Aadhaar number must be unique.
- Mandatory attributes cannot be NULL.

2.2 DISCIPLINE

Represents the broad subject area under which NPTEL courses are grouped (e.g., Computer Science, Electrical Engineering, Management).

Attribute	Type	Constraints
discipline_id	NUMBER	PRIMARY KEY
discipline_name	VARCHAR2(100)	NOT NULL, UNIQUE
description	VARCHAR2(500)	NULL

2.3 INSTITUTION

Represents IITs and IISc that offer courses on NPTEL.

Attribute	Type	Constraints
institution_id	NUMBER	PRIMARY KEY
institution_name	VARCHAR2(150)	NOT NULL, UNIQUE
city	VARCHAR2(100)	NOT NULL
website	VARCHAR2(200)	NULL

2.4 INSTRUCTOR

Represents the IIT/IISc faculty member who designs and delivers an NPTEL course.

Attribute	Type	Constraints
instructor_id	NUMBER	PRIMARY KEY
full_name	VARCHAR2(100)	NOT NULL
designation	VARCHAR2(100)	NOT NULL
email	VARCHAR2(150)	NOT NULL, UNIQUE
phone	VARCHAR2(15)	NOT NULL, UNIQUE
institution_id	NUMBER	NOT NULL, FOREIGN KEY REFERENCES INSTITUTION(institution_id)

2.5 COURSE

Represents an individual NPTEL online course offered in a given semester.

Attribute	Type	Constraints
course_id	NUMBER	PRIMARY KEY
course_name	VARCHAR2(200)	NOT NULL
discipline_id	NUMBER	NOT NULL, FOREIGN KEY REFERENCES DISCIPLINE(discipline_id)
duration_weeks	NUMBER	NOT NULL, CHECK (duration_weeks > 0)
semester	VARCHAR2(20)	NOT NULL, CHECK (IN 'Jan-Apr','Jul-Oct')
year	NUMBER(4)	NOT NULL
total_assignments	NUMBER	NOT NULL, CHECK (> 0)
best_of_n	NUMBER	NOT NULL
exam_date	DATE	NOT NULL
status	VARCHAR2(20)	DEFAULT 'Active', CHECK (IN 'Active','Completed','Cancelled')

2.6 COURSE_INSTRUCTOR

Associative entity resolving the M:N relationship between COURSE and INSTRUCTOR.

Attribute	Type	Constraints
course_id	NUMBER	NOT NULL, FOREIGN KEY REFERENCES COURSE(course_id)
instructor_id	NUMBER	NOT NULL, FOREIGN KEY REFERENCES INSTRUCTOR(instructor_id)
role	VARCHAR2(30)	CHECK (IN 'Primary','Co-instructor')

Constraint: PRIMARY KEY (course_id, instructor_id)

2.7 WEEKLY_ASSIGNMENT

Represents individual weekly quizzes/assignments released for a course.

Attribute	Type	Constraints
assignment_id	NUMBER	PRIMARY KEY
course_id	NUMBER	NOT NULL, FOREIGN KEY REFERENCES COURSE(course_id) ON DELETE CASCADE
week_number	NUMBER	NOT NULL, CHECK (week_number > 0)
release_date	DATE	NOT NULL
due_date	DATE	NOT NULL
max_marks	NUMBER	NOT NULL, CHECK (max_marks > 0)

Constraint: UNIQUE (course_id, week_number). An assignment cannot exist without an associated course.

2.8 ENROLLMENT

Associative entity between LEARNER and COURSE, representing free course enrollment.

Attribute	Type	Constraints
enrollment_id	NUMBER	PRIMARY KEY
learner_id	NUMBER	NOT NULL, FOREIGN KEY REFERENCES LEARNER(learner_id)
course_id	NUMBER	NOT NULL, FOREIGN KEY REFERENCES COURSE(course_id)
enrollment_date	DATE	DEFAULT SYSDATE, NOT NULL
assignment_score	NUMBER	CHECK ($\geq$ 0), computed after course end
status	VARCHAR2(20)	DEFAULT 'Active', CHECK (IN 'Active','Dropped','Completed')

Constraint: UNIQUE (learner_id, course_id). A learner can enroll in a course only once per semester.

2.9 ASSIGNMENT_SUBMISSION

Records each learner's submission for a weekly assignment.

Attribute	Type	Constraints
submission_id	NUMBER	PRIMARY KEY
enrollment_id	NUMBER	NOT NULL, FOREIGN KEY REFERENCES ENROLLMENT(enrollment_id)
assignment_id	NUMBER	NOT NULL, FOREIGN KEY REFERENCES WEEKLY_ASSIGNMENT(assignment_id)
submitted_at	TIMESTAMP	DEFAULT SYSTIMESTAMP, NOT NULL
marks_obtained	NUMBER	NOT NULL, CHECK ($\geq$ 0)

Constraint: UNIQUE (enrollment_id, assignment_id). A learner can submit each weekly assignment only once.

2.10 EXAM_REGISTRATION

Represents a learner opting for the proctored certification exam for an enrolled course.

Attribute	Type	Constraints
exam_reg_id	NUMBER	PRIMARY KEY
enrollment_id	NUMBER	NOT NULL, UNIQUE, FOREIGN KEY REFERENCES ENROLLMENT(enrollment_id)
reg_date	DATE	DEFAULT SYSDATE, NOT NULL
preferred_city	VARCHAR2(100)	NOT NULL
status	VARCHAR2(20)	DEFAULT 'Pending', CHECK (IN 'Pending','Confirmed','Cancelled')

2.11 PAYMENT

Stores proctored exam fee transaction details corresponding to an exam registration.

Attribute	Type	Constraints
payment_id	NUMBER	PRIMARY KEY
exam_reg_id	NUMBER	NOT NULL, UNIQUE, FOREIGN KEY REFERENCES EXAM_REGISTRATION(exam_reg_id)
amount	NUMBER	NOT NULL, CHECK (amount > 0)
payment_date	DATE	DEFAULT SYSDATE
payment_status	VARCHAR2(20)	NOT NULL, CHECK (IN 'Success','Failed','Pending')
transaction_id	VARCHAR2(100)	NOT NULL, UNIQUE
payment_mode	VARCHAR2(30)	NOT NULL, CHECK (IN 'UPI','NetBanking','DebitCard','CreditCard')

2.12 TEST_CENTRE

Represents NPTEL-approved proctored examination centres across India (typically TCS iON centres).

Attribute	Type	Constraints
centre_id	NUMBER	PRIMARY KEY
centre_name	VARCHAR2(200)	NOT NULL
city	VARCHAR2(100)	NOT NULL
state	VARCHAR2(100)	NOT NULL
capacity	NUMBER	NOT NULL, CHECK (capacity > 0)
operator	VARCHAR2(100)	NOT NULL (e.g., TCS iON, Aptech)

2.13 HALL

Represents individual halls within a Test Centre.

Attribute	Type	Constraints
hall_id	NUMBER	PRIMARY KEY
centre_id	NUMBER	NOT NULL, FOREIGN KEY REFERENCES TEST_CENTRE(centre_id)
hall_number	VARCHAR2(10)	NOT NULL
capacity	NUMBER	NOT NULL, CHECK (capacity > 0)

2.14 HALL_ALLOCATION

Assigns confirmed exam registrations to Test Centres, halls, and seat numbers.

Attribute	Type	Constraints
allocation_id	NUMBER	PRIMARY KEY
exam_reg_id	NUMBER	NOT NULL, UNIQUE, FOREIGN KEY REFERENCES EXAM_REGISTRATION(exam_reg_id)
centre_id	NUMBER	NOT NULL, FOREIGN KEY REFERENCES TEST_CENTRE(centre_id)
hall_id	NUMBER	NOT NULL, FOREIGN KEY REFERENCES HALL(hall_id)
seat_number	VARCHAR2(10)	NOT NULL

2.15 EVALUATOR

Represents authorized personnel (IIT/IISc faculty or trained evaluators) who assess proctored answer sheets.

Attribute	Type	Constraints
evaluator_id	NUMBER	PRIMARY KEY
full_name	VARCHAR2(100)	NOT NULL
qualification	VARCHAR2(100)	NOT NULL
email	VARCHAR2(150)	NOT NULL, UNIQUE
phone	VARCHAR2(15)	NOT NULL, UNIQUE

2.16 ANSWER_SHEET

Represents the proctored exam answer scripts submitted by learners for evaluation.

Attribute	Type	Constraints
sheet_id	NUMBER	PRIMARY KEY
exam_reg_id	NUMBER	NOT NULL, UNIQUE, FOREIGN KEY REFERENCES EXAM_REGISTRATION(exam_reg_id)
evaluator_id	NUMBER	NOT NULL, FOREIGN KEY REFERENCES EVALUATOR(evaluator_id)
submission_date	DATE	DEFAULT SYSDATE
marks_obtained	NUMBER	CHECK ($\geq$ 0)
evaluation_status	VARCHAR2(20)	DEFAULT 'Pending', CHECK (IN 'Pending','Evaluated')

2.17 RESULT

Stores the final computed result and certificate category for an enrollment.

Attribute	Type	Constraints
result_id	NUMBER	PRIMARY KEY
enrollment_id	NUMBER	NOT NULL, UNIQUE, FOREIGN KEY REFERENCES ENROLLMENT(enrollment_id)
assignment_score	NUMBER	NOT NULL, CHECK ($\geq$ 0)
exam_score	NUMBER	CHECK ($\geq$ 0) NULL if absent
final_score	NUMBER	NOT NULL (25% assignment + 75% exam)
certificate_type	VARCHAR2(30)	CHECK (IN 'Topper','Elite+Gold','Elite','Pass','No Certificate')
result_date	DATE	DEFAULT SYSDATE

2.18 CERTIFICATE

Records details of the digital certificate issued to qualifying learners.

Attribute	Type	Constraints
certificate_id	NUMBER	PRIMARY KEY
result_id	NUMBER	NOT NULL, UNIQUE, FOREIGN KEY REFERENCES RESULT(result_id)
issue_date	DATE	NOT NULL, DEFAULT SYSDATE
certificate_no	VARCHAR2(50)	NOT NULL, UNIQUE
download_url	VARCHAR2(500)	NOT NULL

2.19 GRIEVANCE

Maintains records of complaints raised by learners after result declaration.

Attribute	Type	Constraints
grievance_id	NUMBER	PRIMARY KEY
enrollment_id	NUMBER	NOT NULL, FOREIGN KEY REFERENCES ENROLLMENT(enrollment_id)
grievance_type	VARCHAR2(50)	NOT NULL, CHECK (IN 'Assignment Score','Exam Score','Certificate','Other')
grievance_text	VARCHAR2(1000)	NOT NULL
grievance_date	DATE	DEFAULT SYSDATE
status	VARCHAR2(20)	DEFAULT 'Open', CHECK (IN 'Open','Resolved','Rejected')
resolved_date	DATE	NULL

Chapter 3

Identification of Relationships in NPTEL

Based on the system analysis, the following relationships are identified in the National Examination Management System (NPTEL). Each relationship is defined with its cardinality, participation constraints, and business rules.

3.1 LEARNER — ENROLLS IN — COURSE

Relationship Name: Enrolls In
Participating Entities: LEARNER and COURSE
Cardinality: M:N
Participation: LEARNER – Partial COURSE – Partial
Business Rule: A learner can enroll in multiple courses. A course can have many learners. Enrollment is free with no fee required.
Resolution: Resolved using associative entity: ENROLLMENT(learner_id, course_id)

3.2 ENROLLMENT — SUBMITS — ASSIGNMENT_SUBMISSION

Relationship Name: Submits Assignment
Participating Entities: ENROLLMENT and WEEKLY_ASSIGNMENT via ASSIGNMENT_SUBMISSION
Cardinality: M:N (resolved via ASSIGNMENT_SUBMISSION)
Participation: ASSIGNMENT_SUBMISSION – Total ENROLLMENT – Partial
Business Rule: Each enrolled learner may submit each weekly assignment once. Unsubmitted assignments score zero.

3.3 COURSE — HAS — WEEKLY_ASSIGNMENT

Relationship Name: Has Assignment
Participating Entities: COURSE and WEEKLY_ASSIGNMENT
Cardinality: 1:N
Participation: WEEKLY_ASSIGNMENT – Total COURSE – Partial
Business Rule: One course has multiple weekly assignments. Each assignment belongs to exactly one course.

3.4 COURSE — BELONGS TO — DISCIPLINE

Relationship Name: Belongs To
Participating Entities: COURSE and DISCIPLINE
Cardinality: N:1

Participation: COURSE – Total DISCIPLINE – Partial
Business Rule: Each course is categorized under exactly one discipline. A discipline contains many courses.

3.5 COURSE — TAUGHT BY — INSTRUCTOR

Relationship Name: Taught By
Participating Entities: COURSE and INSTRUCTOR
Cardinality: M:N
Participation: Partial on both sides
Business Rule: An NPTEL course may have one or more instructors (primary + co-instructors). An instructor may teach multiple courses.
Resolution: Resolved using associative entity: COURSE_INSTRUCTOR(course_id, instructor_id)

3.6 INSTRUCTOR — BELONGS TO — INSTITUTION

Relationship Name: Belongs To
Participating Entities: INSTRUCTOR and INSTITUTION
Cardinality: N:1
Participation: INSTRUCTOR – Total INSTITUTION – Partial
Business Rule: Each instructor is affiliated with exactly one IIT/IISc. An institution has many instructors.

3.7 ENROLLMENT — REGISTERS FOR — EXAM_REGISTRATION

Relationship Name: Registers For Exam
Participating Entities: ENROLLMENT and EXAM_REGISTRATION
Cardinality: 1:1
Participation: ENROLLMENT – Partial EXAM_REGISTRATION – Total
Business Rule: A learner must be enrolled in a course before registering for its proctored exam. Each enrollment can have at most one exam registration.

3.8 EXAM_REGISTRATION — HAS — PAYMENT

Relationship Name: Has Payment
Participating Entities: EXAM_REGISTRATION and PAYMENT
Cardinality: 1:1
Participation: Total on both sides
Business Rule: Each exam registration must have exactly one payment record. A payment record corresponds to exactly one exam registration.

3.9 EXAM_REGISTRATION — ALLOCATED TO — TEST_CENTRE

Relationship Name: Allocated To
Participating Entities: EXAM_REGISTRATION and TEST_CENTRE
Cardinality: N:1

Participation: EXAM_REGISTRATION – Total (after confirmation)   TEST_CENTRE – Partial
Business Rule: Each confirmed exam registration is allocated to exactly one Test Centre. A centre accommodates many learners based on capacity.

3.10 TEST_CENTRE — CONTAINS — HALL

Relationship Name: Contains
Participating Entities: TEST_CENTRE and HALL
Cardinality: 1:N
Participation: HALL – Total   TEST_CENTRE – Partial
Business Rule: Each Test Centre contains one or more halls. Each hall belongs to exactly one centre.

3.11 EXAM_REGISTRATION — ASSIGNED SEAT — HALL_ALLOCATION

Relationship Name: Assigned Seat
Participating Entities: EXAM_REGISTRATION and HALL_ALLOCATION
Cardinality: 1:1
Participation: Total on both sides
Business Rule: Each confirmed exam registration receives exactly one hall and seat allocation.

3.12 EXAM_REGISTRATION — SUBMITS — ANSWER_SHEET

Relationship Name: Submits
Participating Entities: EXAM_REGISTRATION and ANSWER_SHEET
Cardinality: 1:1
Participation: ANSWER_SHEET – Total   EXAM_REGISTRATION – Partial
Business Rule: Each exam registration generates exactly one answer sheet for proctored evaluation.

3.13 EVALUATOR — EVALUATES — ANSWER_SHEET

Relationship Name: Evaluates
Participating Entities: EVALUATOR and ANSWER_SHEET
Cardinality: 1:N
Participation: ANSWER_SHEET – Total   EVALUATOR – Partial
Business Rule: One evaluator can evaluate multiple answer sheets. Each answer sheet is evaluated by exactly one evaluator.

3.14 ENROLLMENT — GENERATES — RESULT

Relationship Name: Generates Result
Participating Entities: ENROLLMENT and RESULT
Cardinality: 1:1
Participation: RESULT – Total   ENROLLMENT – Partial

Business Rule: Each enrollment generates exactly one result record after the course and evaluation are complete.

3.15 RESULT — ISSUES — CERTIFICATE

Relationship Name: Issues Certificate
Participating Entities: RESULT and CERTIFICATE
Cardinality: 1:1
Participation: CERTIFICATE – Total RESULT – Partial
Business Rule: A certificate is issued only if the certificate_type is one of: Topper, Elite+Gold, Elite, or Pass.

3.16 ENROLLMENT — RAISES — GRIEVANCE

Relationship Name: Raises Grievance
Participating Entities: ENROLLMENT and GRIEVANCE
Cardinality: 1:N
Participation: GRIEVANCE – Total ENROLLMENT – Partial
Business Rule: A learner may raise multiple grievances related to one course enrollment. Grievances are raised only after result publication.

Chapter 4

ER Diagram for NPTEL

4.1 Overview

This chapter presents the Entity-Relationship (ER) Diagram for the National Examination Management System (NPTEL). The ER diagram provides a complete visual representation of all entities, attributes, relationships, and cardinality constraints identified in the preceding chapters.

The diagram illustrates how the 19 entities of the NPTEL system are interconnected, covering the full lifecycle from learner registration and course enrollment through proctored examination management to certificate issuance and grievance resolution.

4.2 ER Diagram

The interactive ER Diagram for the NPTEL system has been modelled and is available as an HTML file. Click the link below to open the diagram in your web browser:

Click Here to View the ER Diagram (NPTEL)

Note: Ensure that the file `ER_diagram.html` is placed in the same directory as this PDF document. Clicking the link above will open the interactive ER diagram directly in your default web browser.

4.3 Key Entities Represented

The ER diagram includes the following entities and their relationships:

- **LEARNER** – Registers on the SWAYAM/NPTEL portal and enrolls in courses.
- **DISCIPLINE** – Broad subject category under which courses are grouped.
- **INSTITUTION** – IIT/IISc that offers NPTEL courses.
- **INSTRUCTOR** – Faculty member affiliated with an IIT/IISc who delivers a course.
- **COURSE** – Individual NPTEL online course offered per semester.
- **COURSE_INSTRUCTOR** – Associative entity resolving the M:N relationship between COURSE and INSTRUCTOR.
- **WEEKLY_ASSIGNMENT** – Weekly quiz/assignment released for a course.
- **ENROLLMENT** – Free course enrollment linking LEARNER and COURSE.
- **ASSIGNMENT_SUBMISSION** – Individual weekly assignment submission record.
- **EXAM_REGISTRATION** – Learner opt-in for the proctored certification exam.

- **PAYMENT** – Proctored exam fee transaction record.
- **TEST_CENTRE** – NPTEL-approved proctored examination centre.
- **HALL** – Individual hall within a Test Centre.
- **HALL_ALLOCATION** – Seat and hall assignment for a confirmed exam registration.
- **EVALUATOR** – Authorized evaluator for proctored answer sheets.
- **ANSWER_SHEET** – Proctored exam answer script submitted for evaluation.
- **RESULT** – Final computed result with certificate category.
- **CERTIFICATE** – Digital certificate issued to qualifying learners.
- **GRIEVANCE** – Learner complaint raised after result declaration.

Chapter 5

Conversion of ER Diagram to Relational Schema

Based on the ER model of the National Examination Management System (NPTEL), the following relational schema is derived using standard ER-to-Relational mapping rules.

- Each strong entity is converted into a separate relation.
- Weak entities include foreign keys referencing their owner entities.
- M:N relationships are resolved using associative relations.
- Primary keys and foreign keys enforce entity integrity and referential integrity.

5.1 Strong Entity Relations

5.1.1 LEARNER

Stores registered learner information. Primary Key: `learner_id`

```
LEARNER (
  learner_id          NUMBER PRIMARY KEY,
  full_name           VARCHAR2(100) NOT NULL,
  dob                 DATE NOT NULL,
  gender              VARCHAR2(10)
                      CHECK (gender IN ('Male','Female','Other'))
   NOT NULL,
  email               VARCHAR2(150) UNIQUE NOT NULL,
  phone               VARCHAR2(15) UNIQUE NOT NULL,
  aadhaar_no          VARCHAR2(12) UNIQUE NOT NULL,
  education_level     VARCHAR2(50) NOT NULL,
  institution_name    VARCHAR2(200) NOT NULL,
  state_of_residence  VARCHAR2(100) NOT NULL,
  password_hash       VARCHAR2(255) NOT NULL,
  registration_date   DATE DEFAULT SYSDATE NOT NULL
);
```

All attributes are atomic and functionally dependent on `learner_id`.

5.1.2 DISCIPLINE

Stores NPTEL course disciplines. Primary Key: `discipline_id`

```
DISCIPLINE (
  discipline_id   NUMBER PRIMARY KEY,
  discipline_name VARCHAR2(100) UNIQUE NOT NULL,
```

```
  description     VARCHAR2(500)
);
```

5.1.3 INSTITUTION

Stores IIT/IISc records. Primary Key: `institution_id`

```
INSTITUTION (
  institution_id   NUMBER PRIMARY KEY,
  institution_name VARCHAR2(150) UNIQUE NOT NULL,
  city             VARCHAR2(100) NOT NULL,
  website          VARCHAR2(200)
);
```

5.1.4 INSTRUCTOR

Stores NPTEL faculty details. Primary Key: `instructor_id` Foreign Key: `institution_id` references INSTITUTION

```
INSTRUCTOR (
  instructor_id  NUMBER PRIMARY KEY,
  full_name      VARCHAR2(100) NOT NULL,
  designation    VARCHAR2(100) NOT NULL,
  email          VARCHAR2(150) UNIQUE NOT NULL,
  phone          VARCHAR2(15) UNIQUE NOT NULL,
  institution_id NUMBER NOT NULL,
  FOREIGN KEY (institution_id) REFERENCES INSTITUTION(
   institution_id)
);
```

5.1.5 COURSE

Represents an NPTEL online course per semester. Primary Key: `course_id` Foreign Key: `discipline_id` references DISCIPLINE

```
COURSE (
  course_id         NUMBER PRIMARY KEY,
  course_name       VARCHAR2(200) NOT NULL,
  discipline_id     NUMBER NOT NULL,
  duration_weeks    NUMBER CHECK (duration_weeks > 0) NOT NULL,
  semester          VARCHAR2(20)
                    CHECK (semester IN ('Jan-Apr','Jul-Oct')) NOT
   NULL,
  year              NUMBER(4) NOT NULL,
  total_assignments NUMBER NOT NULL,
  best_of_n         NUMBER NOT NULL,
  exam_date         DATE NOT NULL,
  status            VARCHAR2(20) DEFAULT 'Active'
                    CHECK (status IN ('Active','Completed','
   Cancelled')),
```

```
  FOREIGN KEY (discipline_id) REFERENCES DISCIPLINE(discipline_id)
);
```

5.1.6 TEST_CENTRE

Stores NPTEL-approved proctored Test Centre details. Primary Key: `centre_id`

```
TEST_CENTRE (
  centre_id   NUMBER PRIMARY KEY,
  centre_name VARCHAR2(200) NOT NULL,
  city        VARCHAR2(100) NOT NULL,
  state       VARCHAR2(100) NOT NULL,
  capacity    NUMBER CHECK (capacity > 0) NOT NULL,
  operator    VARCHAR2(100) NOT NULL
);
```

5.1.7 HALL

Represents halls within a Test Centre. Primary Key: `hall_id` Foreign Key: `centre_id` references TEST_CENTRE

```
HALL (
  hall_id     NUMBER PRIMARY KEY,
  centre_id   NUMBER NOT NULL,
  hall_number VARCHAR2(10) NOT NULL,
  capacity    NUMBER CHECK (capacity > 0) NOT NULL,
  FOREIGN KEY (centre_id) REFERENCES TEST_CENTRE(centre_id)
);
```

5.1.8 EVALUATOR

Stores authorized evaluator details. Primary Key: `evaluator_id`

```
EVALUATOR (
  evaluator_id  NUMBER PRIMARY KEY,
  full_name     VARCHAR2(100) NOT NULL,
  qualification VARCHAR2(100) NOT NULL,
  email         VARCHAR2(150) UNIQUE NOT NULL,
  phone         VARCHAR2(15) UNIQUE NOT NULL
);
```

5.2 Associative and Dependent Relations

5.2.1 COURSE_INSTRUCTOR

Resolves M:N relationship between COURSE and INSTRUCTOR. Primary Key: `(course_id, instructor_id)`

```
COURSE_INSTRUCTOR (
```

```
  course_id     NUMBER NOT NULL,
  instructor_id NUMBER NOT NULL,
  role          VARCHAR2(30) CHECK (role IN ('Primary','Co-
   instructor')),
  PRIMARY KEY (course_id, instructor_id),
  FOREIGN KEY (course_id) REFERENCES COURSE(course_id),
  FOREIGN KEY (instructor_id) REFERENCES INSTRUCTOR(instructor_id)
);
```

5.2.2 WEEKLY_ASSIGNMENT

Dependent on COURSE. Primary Key: `assignment_id`

```
WEEKLY_ASSIGNMENT (
  assignment_id NUMBER PRIMARY KEY,
  course_id     NUMBER NOT NULL,
  week_number   NUMBER NOT NULL,
  release_date  DATE NOT NULL,
  due_date      DATE NOT NULL,
  max_marks     NUMBER CHECK (max_marks > 0) NOT NULL,
  UNIQUE (course_id, week_number),
  FOREIGN KEY (course_id) REFERENCES COURSE(course_id)
    ON DELETE CASCADE
);
```

5.2.3 ENROLLMENT

Resolves M:N between LEARNER and COURSE. Primary Key: `enrollment_id`

```
ENROLLMENT (
  enrollment_id    NUMBER PRIMARY KEY,
  learner_id       NUMBER NOT NULL,
  course_id        NUMBER NOT NULL,
  enrollment_date  DATE DEFAULT SYSDATE NOT NULL,
  assignment_score NUMBER CHECK (assignment_score >= 0),
  status           VARCHAR2(20) DEFAULT 'Active'
                   CHECK (status IN ('Active','Dropped','Completed'
   )),
  UNIQUE (learner_id, course_id),
  FOREIGN KEY (learner_id) REFERENCES LEARNER(learner_id),
  FOREIGN KEY (course_id) REFERENCES COURSE(course_id)
);
```

5.2.4 ASSIGNMENT_SUBMISSION

Records individual weekly assignment submissions.

```
ASSIGNMENT_SUBMISSION (
  submission_id  NUMBER PRIMARY KEY,
  enrollment_id  NUMBER NOT NULL,
```

```
  assignment_id  NUMBER NOT NULL,
  submitted_at   TIMESTAMP DEFAULT SYSTIMESTAMP NOT NULL,
  marks_obtained NUMBER NOT NULL CHECK (marks_obtained >= 0),
  UNIQUE (enrollment_id, assignment_id),
  FOREIGN KEY (enrollment_id) REFERENCES ENROLLMENT(enrollment_id),
  FOREIGN KEY (assignment_id)
    REFERENCES WEEKLY_ASSIGNMENT(assignment_id)
);
```

5.2.5 EXAM_REGISTRATION

Represents proctored exam opt-in. Primary Key: `exam_reg_id`

```
EXAM_REGISTRATION (
  exam_reg_id     NUMBER PRIMARY KEY,
  enrollment_id   NUMBER UNIQUE NOT NULL,
  reg_date        DATE DEFAULT SYSDATE NOT NULL,
  preferred_city  VARCHAR2(100) NOT NULL,
  status          VARCHAR2(20) DEFAULT 'Pending'
                  CHECK (status IN ('Pending','Confirmed','Cancelled
   ')),
  FOREIGN KEY (enrollment_id) REFERENCES ENROLLMENT(enrollment_id)
);
```

5.2.6 PAYMENT

Implements 1:1 relationship with EXAM_REGISTRATION.

```
PAYMENT (
  payment_id      NUMBER PRIMARY KEY,
  exam_reg_id     NUMBER UNIQUE NOT NULL,
  amount          NUMBER CHECK (amount > 0) NOT NULL,
  payment_date    DATE DEFAULT SYSDATE,
  payment_status  VARCHAR2(20)
                  CHECK (payment_status IN
                  ('Success','Failed','Pending')) NOT NULL,
  transaction_id  VARCHAR2(100) UNIQUE NOT NULL,
  payment_mode    VARCHAR2(30)
                  CHECK (payment_mode IN
                  ('UPI','NetBanking','DebitCard','CreditCard')) NOT
    NULL,
  FOREIGN KEY (exam_reg_id)
    REFERENCES EXAM_REGISTRATION(exam_reg_id)
);
```

5.2.7 HALL_ALLOCATION

Assigns confirmed registrations to halls and seats.

```
HALL_ALLOCATION (
```

```
  allocation_id NUMBER PRIMARY KEY,
  exam_reg_id   NUMBER UNIQUE NOT NULL,
  centre_id     NUMBER NOT NULL,
  hall_id       NUMBER NOT NULL,
  seat_number   VARCHAR2(10) NOT NULL,
  FOREIGN KEY (exam_reg_id)
    REFERENCES EXAM_REGISTRATION(exam_reg_id),
  FOREIGN KEY (centre_id) REFERENCES TEST_CENTRE(centre_id),
  FOREIGN KEY (hall_id) REFERENCES HALL(hall_id)
);
```

5.2.8 ANSWER_SHEET

Stores proctored exam evaluation details.

```
ANSWER_SHEET (
  sheet_id            NUMBER PRIMARY KEY,
  exam_reg_id         NUMBER UNIQUE NOT NULL,
  evaluator_id        NUMBER NOT NULL,
  submission_date     DATE DEFAULT SYSDATE,
  marks_obtained      NUMBER CHECK (marks_obtained >= 0),
  evaluation_status VARCHAR2(20) DEFAULT 'Pending'
                      CHECK (evaluation_status IN
                      ('Pending','Evaluated')),
  FOREIGN KEY (exam_reg_id)
    REFERENCES EXAM_REGISTRATION(exam_reg_id),
  FOREIGN KEY (evaluator_id) REFERENCES EVALUATOR(evaluator_id)
);
```

5.2.9 RESULT

Stores final computed results with NPTEL certificate categories.

```
RESULT (
  result_id          NUMBER PRIMARY KEY,
  enrollment_id      NUMBER UNIQUE NOT NULL,
  assignment_score NUMBER NOT NULL CHECK (assignment_score >= 0),
  exam_score         NUMBER CHECK (exam_score >= 0),
  final_score        NUMBER NOT NULL,
  certificate_type VARCHAR2(30)
                     CHECK (certificate_type IN
                     ('Topper','Elite+Gold','Elite',
                      'Pass','No Certificate')),
  result_date        DATE DEFAULT SYSDATE,
  FOREIGN KEY (enrollment_id) REFERENCES ENROLLMENT(enrollment_id)
);
```

5.2.10 CERTIFICATE

Stores digital certificate details issued to qualifying learners.

```sql
CERTIFICATE (
  certificate_id  NUMBER PRIMARY KEY,
  result_id       NUMBER UNIQUE NOT NULL,
  issue_date      DATE NOT NULL DEFAULT SYSDATE,
  certificate_no  VARCHAR2(50) UNIQUE NOT NULL,
  download_url    VARCHAR2(500) NOT NULL,
  FOREIGN KEY (result_id) REFERENCES RESULT(result_id)
);
```

5.2.11 GRIEVANCE

Stores learner grievances raised after result declaration.

```sql
GRIEVANCE (
  grievance_id   NUMBER PRIMARY KEY,
  enrollment_id  NUMBER NOT NULL,
  grievance_type VARCHAR2(50)
                 CHECK (grievance_type IN
                 ('Assignment Score','Exam Score',
                  'Certificate','Other')) NOT NULL,
  grievance_text VARCHAR2(1000) NOT NULL,
  grievance_date DATE DEFAULT SYSDATE,
  status         VARCHAR2(20) DEFAULT 'Open'
                 CHECK (status IN ('Open','Resolved','Rejected')),
  resolved_date  DATE,
  FOREIGN KEY (enrollment_id) REFERENCES ENROLLMENT(enrollment_id)
);
```

5.3 Relational Schema Diagram (ERDPlus)

The following diagram represents the relational schema of the National Examination Management System (NPTEL) as modelled using ERDPlus, showing all tables, primary keys, foreign keys, and their inter-table relationships.

[Relational Schema Diagram – National Examination Management System (NPTEL)]

Chapter 6

Functional Dependencies and Key Identification

Functional Dependencies (FDs) are derived from the attribute constraints and key definitions of each relation in the National Examination Management System (NPTEL). A functional dependency $X \rightarrow Y$ indicates that attribute set X uniquely determines attribute set Y.

For each relation, the primary key functionally determines all non-key attributes. Candidate keys, superkeys, and normalization properties are identified accordingly.

6.1 LEARNER

Relation:
`LEARNER(learner_id, full_name, dob, gender, email, phone, aadhaar_no, education_level, institution_name, state_of_residence, password_hash, registration_date)`

Functional Dependencies:

$$\begin{aligned} \text{learner\_id} &\rightarrow \{\text{full\_name, dob, gender, email, phone, aadhaar\_no,} \\ &\quad \text{education\_level, institution\_name, state\_of\_residence,} \\ &\quad \text{password\_hash, registration\_date}\} \\ \text{email} &\rightarrow \text{learner\_id} \\ \text{phone} &\rightarrow \text{learner\_id} \\ \text{aadhaar\_no} &\rightarrow \text{learner\_id} \end{aligned}$$

Explanation: The `learner_id` uniquely identifies every learner. Due to UNIQUE constraints, email, phone, and Aadhaar number also uniquely determine the learner.

- **Candidate Keys:** learner_id, email, phone, aadhaar_no
- **Primary Key:** learner_id
- **Normalization:** Relation satisfies BCNF as every determinant is a candidate key.

Minimal FDs (Final):

$$
\begin{aligned}
\text{learner\_id} &\rightarrow \text{full\_name}\\
\text{learner\_id} &\rightarrow \text{dob}\\
\text{learner\_id} &\rightarrow \text{gender}\\
\text{learner\_id} &\rightarrow \text{email}\\
\text{learner\_id} &\rightarrow \text{phone}\\
\text{learner\_id} &\rightarrow \text{aadhaar\_no}\\
\text{learner\_id} &\rightarrow \text{education\_level}\\
\text{learner\_id} &\rightarrow \text{institution\_name}\\
\text{learner\_id} &\rightarrow \text{state\_of\_residence}\\
\text{learner\_id} &\rightarrow \text{password\_hash}\\
\text{learner\_id} &\rightarrow \text{registration\_date}\\
\text{email} &\rightarrow \text{learner\_id}\\
\text{phone} &\rightarrow \text{learner\_id}\\
\text{aadhaar\_no} &\rightarrow \text{learner\_id}
\end{aligned}
$$

6.2 DISCIPLINE

Relation: `DISCIPLINE(discipline_id, discipline_name, description)`

$$
\begin{aligned}
\text{discipline\_id} &\rightarrow \{\text{discipline\_name, description}\}\\
\text{discipline\_name} &\rightarrow \text{discipline\_id}
\end{aligned}
$$

- **Candidate Keys:** discipline_id, discipline_name
- **Primary Key:** discipline_id
- **Normalization:** BCNF satisfied.

Minimal FDs (Final):

$$
\begin{aligned}
\text{discipline\_id} &\rightarrow \text{discipline\_name}\\
\text{discipline\_id} &\rightarrow \text{description}\\
\text{discipline\_name} &\rightarrow \text{discipline\_id}
\end{aligned}
$$

6.3 INSTITUTION

Relation: `INSTITUTION(institution_id, institution_name, city, website)`

$$
\begin{aligned}
\text{institution\_id} &\rightarrow \{\text{institution\_name, city, website}\}\\
\text{institution\_name} &\rightarrow \text{institution\_id}
\end{aligned}
$$

- **Candidate Keys:** institution_id, institution_name

- **Primary Key:** institution_id
- **Normalization:** BCNF satisfied.

Minimal FDs (Final):

$$\begin{aligned} \text{institution\_id} &\rightarrow \text{institution\_name} \\ \text{institution\_id} &\rightarrow \text{city} \\ \text{institution\_id} &\rightarrow \text{website} \\ \text{institution\_name} &\rightarrow \text{institution\_id} \end{aligned}$$

6.4 INSTRUCTOR

Relation:
`INSTRUCTOR(instructor_id, full_name, designation, email, phone, institutio`

$$\begin{aligned} \text{instructor\_id} &\rightarrow \{\text{full\_name, designation, email, phone, institution\_id}\} \\ \text{email} &\rightarrow \text{instructor\_id} \\ \text{phone} &\rightarrow \text{instructor\_id} \end{aligned}$$

- **Candidate Keys:** instructor_id, email, phone
- **Primary Key:** instructor_id
- **Normalization:** BCNF satisfied.

Minimal FDs (Final):

$$\begin{aligned} \text{instructor\_id} &\rightarrow \text{full\_name} \\ \text{instructor\_id} &\rightarrow \text{designation} \\ \text{instructor\_id} &\rightarrow \text{email} \\ \text{instructor\_id} &\rightarrow \text{phone} \\ \text{instructor\_id} &\rightarrow \text{institution\_id} \\ \text{email} &\rightarrow \text{instructor\_id} \\ \text{phone} &\rightarrow \text{instructor\_id} \end{aligned}$$

6.5 COURSE

Relation:
`COURSE(course_id, course_name, discipline_id, duration_weeks, semester, year, total_assignments, best_of_n, exam_date, status)`

$$\begin{aligned} \text{course\_id} \rightarrow \{&\text{course\_name, discipline\_id, duration\_weeks, semester,} \\ &\text{year, total\_assignments, best\_of\_n, exam\_date, status}\} \end{aligned}$$

Explanation: No alternate keys exist since course names may repeat across semesters.

- **Primary Key:** course_id
- **Normalization:** BCNF satisfied.

Minimal FDs (Final):

$$\begin{aligned}
\text{course_id} &\rightarrow \text{course_name} \\
\text{course_id} &\rightarrow \text{discipline_id} \\
\text{course_id} &\rightarrow \text{duration_weeks} \\
\text{course_id} &\rightarrow \text{semester} \\
\text{course_id} &\rightarrow \text{year} \\
\text{course_id} &\rightarrow \text{total_assignments} \\
\text{course_id} &\rightarrow \text{best_of_n} \\
\text{course_id} &\rightarrow \text{exam_date} \\
\text{course_id} &\rightarrow \text{status}
\end{aligned}$$

6.6 WEEKLY ASSIGNMENT

Relation:
`WEEKLY_ASSIGNMENT(assignment_id, course_id, week_number, release_date, due_date, max_marks)`

$$\text{assignment\_id} \rightarrow \{\text{course\_id, week\_number, release\_date, due\_date, max\_marks}\}$$
$$(\text{course\_id},\ \text{week\_number}) \rightarrow \text{assignment\_id}$$

- **Candidate Keys:** assignment_id, (course_id, week_number)
- **Primary Key:** assignment_id
- **Normalization:** BCNF satisfied.

Minimal FDs (Final):

$$\begin{aligned}
\text{assignment_id} &\rightarrow \text{course_id} \\
\text{assignment_id} &\rightarrow \text{week_number} \\
\text{assignment_id} &\rightarrow \text{release_date} \\
\text{assignment_id} &\rightarrow \text{due_date} \\
\text{assignment_id} &\rightarrow \text{max_marks} \\
(\text{course_id},\ \text{week_number}) &\rightarrow \text{assignment_id}
\end{aligned}$$

6.7 ENROLLMENT

Relation:
`ENROLLMENT(enrollment_id, learner_id, course_id, enrollment_date, assignment_score, status)`

$$\text{enrollment\_id} \rightarrow \{\text{learner\_id, course\_id, enrollment\_date, assignment\_score, status}\}$$
$$(\text{learner\_id},\ \text{course\_id}) \rightarrow \text{enrollment\_id}$$

Explanation: Composite key determines `enrollment_id`, which determines all remaining attributes.

- **Candidate Keys:** enrollment_id, (learner_id, course_id)
- **Primary Key:** enrollment_id
- **Normalization:** BCNF satisfied.

Minimal FDs (Final):

$$\begin{aligned}
\text{enrollment_id} &\rightarrow \text{learner_id}\\
\text{enrollment_id} &\rightarrow \text{course_id}\\
\text{enrollment_id} &\rightarrow \text{enrollment_date}\\
\text{enrollment_id} &\rightarrow \text{assignment_score}\\
\text{enrollment_id} &\rightarrow \text{status}\\
(\text{learner_id},\ \text{course_id}) &\rightarrow \text{enrollment_id}
\end{aligned}$$

6.8 ASSIGNMENT SUBMISSION

Relation:
`ASSIGNMENT_SUBMISSION(submission_id, enrollment_id, assignment_id, submitted_at, marks_obtained)`

$$\text{submission\_id} \rightarrow \{\text{enrollment\_id, assignment\_id, submitted\_at, marks\_obtained}\}$$
$$(\text{enrollment\_id},\ \text{assignment\_id}) \rightarrow \text{submission\_id}$$

- **Candidate Keys:** submission_id, (enrollment_id, assignment_id)
- **Primary Key:** submission_id
- **Normalization:** BCNF satisfied.

Minimal FDs (Final):

$$\begin{aligned}
\text{submission_id} &\rightarrow \text{enrollment_id}\\
\text{submission_id} &\rightarrow \text{assignment_id}\\
\text{submission_id} &\rightarrow \text{submitted_at}\\
\text{submission_id} &\rightarrow \text{marks_obtained}\\
(\text{enrollment_id},\ \text{assignment_id}) &\rightarrow \text{submission_id}
\end{aligned}$$

6.9 EXAM REGISTRATION

Relation:
`EXAM_REGISTRATION(exam_reg_id, enrollment_id, reg_date, preferred_city, status)`

$$\text{exam\_reg\_id} \rightarrow \{\text{enrollment\_id, reg\_date, preferred\_city, status}\}$$
$$\text{enrollment\_id} \rightarrow \text{exam\_reg\_id}$$

- **Candidate Keys:** exam_reg_id, enrollment_id
- **Primary Key:** exam_reg_id
- **Normalization:** BCNF satisfied.

Minimal FDs (Final):

$$\begin{aligned}
\text{exam_reg_id} &\to \text{enrollment_id}\\
\text{exam_reg_id} &\to \text{reg_date}\\
\text{exam_reg_id} &\to \text{preferred_city}\\
\text{exam_reg_id} &\to \text{status}\\
\text{enrollment_id} &\to \text{exam_reg_id}
\end{aligned}$$

6.10 PAYMENT

Relation:
`PAYMENT(payment_id, exam_reg_id, amount, payment_date, payment_status, transaction_id, payment_mode)`

$$\begin{aligned}
\text{payment_id} &\to \{\text{exam_reg_id, amount, payment_date, payment_status, transaction_id, payment_mode}\}\\
\text{exam_reg_id} &\to \text{payment_id}\\
\text{transaction_id} &\to \text{payment_id}
\end{aligned}$$

- **Candidate Keys:** payment_id, exam_reg_id, transaction_id
- **Primary Key:** payment_id
- **Normalization:** BCNF satisfied.

Minimal FDs (Final):

$$\begin{aligned}
\text{payment_id} &\to \text{exam_reg_id}\\
\text{payment_id} &\to \text{amount}\\
\text{payment_id} &\to \text{payment_date}\\
\text{payment_id} &\to \text{payment_status}\\
\text{payment_id} &\to \text{transaction_id}\\
\text{payment_id} &\to \text{payment_mode}\\
\text{exam_reg_id} &\to \text{payment_id}\\
\text{transaction_id} &\to \text{payment_id}
\end{aligned}$$

6.11 COURSE INSTRUCTOR

Relation: `COURSE_INSTRUCTOR(course_id, instructor_id, role)`

$$(\text{course\_id},\ \text{instructor\_id}) \to \text{role}$$

- **Primary Key:** (course_id, instructor_id)

- **Normalization:** BCNF satisfied.

Minimal FDs (Final):

$$(\text{course\_id},\ \text{instructor\_id}) \rightarrow \text{role}$$

6.12 RESULT

Relation:
`RESULT(result_id, enrollment_id, assignment_score, exam_score, final_score, certificate_type, result_date)`

$$\text{result\_id} \rightarrow \{\text{enrollment\_id, assignment\_score, exam\_score, final\_score, certificate\_type, result\_date}\}$$
$$\text{enrollment\_id} \rightarrow \text{result\_id}$$

Explanation: `enrollment_id` determines `result_id`, which determines all other attributes.

- **Candidate Keys:** result_id, enrollment_id
- **Primary Key:** result_id
- **Normalization:** BCNF satisfied.

Minimal FDs (Final):

$$\begin{aligned}
\text{result_id} &\rightarrow \text{enrollment_id}\\
\text{result_id} &\rightarrow \text{assignment_score}\\
\text{result_id} &\rightarrow \text{exam_score}\\
\text{result_id} &\rightarrow \text{final_score}\\
\text{result_id} &\rightarrow \text{certificate_type}\\
\text{result_id} &\rightarrow \text{result_date}\\
\text{enrollment_id} &\rightarrow \text{result_id}
\end{aligned}$$

6.13 CERTIFICATE

Relation:
`CERTIFICATE(certificate_id, result_id, issue_date, certificate_no, download_url)`

$$\begin{aligned}
\text{certificate_id} &\rightarrow \{\text{result_id, issue_date, certificate_no, download_url}\}\\
\text{result_id} &\rightarrow \text{certificate_id}\\
\text{certificate_no} &\rightarrow \text{certificate_id}
\end{aligned}$$

- **Candidate Keys:** certificate_id, result_id, certificate_no
- **Primary Key:** certificate_id
- **Normalization:** BCNF satisfied.

Minimal FDs (Final):

$$\begin{aligned}
\text{certificate_id} &\rightarrow \text{result_id}\\
\text{certificate_id} &\rightarrow \text{issue_date}\\
\text{certificate_id} &\rightarrow \text{certificate_no}\\
\text{certificate_id} &\rightarrow \text{download_url}\\
\text{result_id} &\rightarrow \text{certificate_id}\\
\text{certificate_no} &\rightarrow \text{certificate_id}
\end{aligned}$$

6.14 GRIEVANCE

Relation:
`GRIEVANCE(grievance_id, enrollment_id, grievance_type, grievance_text, grievance_date, status, resolved_date)`

grievance_id → {enrollment_id, grievance_type, grievance_text, grievance_date, status, resolved_date}

Explanation: Multiple grievances per enrollment are allowed, so `enrollment_id` alone is not a candidate key.

- **Primary Key:** grievance_id
- **Normalization:** BCNF satisfied.

Minimal FDs (Final):

$$\begin{aligned}
\text{grievance_id} &\rightarrow \text{enrollment_id}\\
\text{grievance_id} &\rightarrow \text{grievance_type}\\
\text{grievance_id} &\rightarrow \text{grievance_text}\\
\text{grievance_id} &\rightarrow \text{grievance_date}\\
\text{grievance_id} &\rightarrow \text{status}\\
\text{grievance_id} &\rightarrow \text{resolved_date}
\end{aligned}$$

Overall Observations

- All strong entities satisfy BCNF since every determinant is a candidate key.
- 1:1 relationships introduce alternate candidate keys.
- M:N relationships use composite primary keys.
- No partial or transitive dependencies exist.
- The overall schema satisfies 3NF and BCNF.

Chapter 7

Database Normalization

7.1 Step 0: Unnormalized Relation (UNF)

From the initial conceptual design of the National Examination Management System (NPTEL), assume that all attributes are combined into a single universal relation:

```
R(learner_id, full_name, email, dob, phone_numbers,
  course_id, course_name, exam_date, discipline_name,
  instructor_name, institution_name,
  week_number, max_marks, assignment_score,
  centre_id, centre_name, city, state,
  payment_id, amount, evaluator_id, marks_obtained,
  certificate_type, grievance_id, grievance_type, status)
```

Issues Identified

Several problems exist in this unnormalized structure:

- `phone_numbers` may be multivalued – a learner may have multiple phone numbers.
- Repeating groups exist – a learner may enroll in multiple courses, each with multiple weekly assignments.
- Course information depends only on `course_id`.
- Centre information depends only on `centre_id`.
- Instructor information depends only on `instructor_id`.
- Payment information depends only on `payment_id`.
- Grievance information depends only on `grievance_id`.

Thus, the relation contains multivalued attributes, repeating groups, partial dependencies, and transitive dependencies. Therefore, the relation does not satisfy First Normal Form (1NF).

7.2 Step 1: First Normal Form (1NF)

Rule: The domain of an attribute must include only atomic (simple, indivisible) values.

7.2.1 Relation and Functional Dependencies

Relation: STUDENT(student_id, full_name, dob, gender, phone_numbers)

Functional Dependencies:

$$\text{FD1:}\quad \texttt{student\_id} \rightarrow \texttt{full\_name,}\ \ \texttt{dob,}\ \ \texttt{gender}$$

Note: `phone_numbers` is a multivalued attribute, which violates relational atomicity before FDs are fully applied.

7.2.2 Closure to Find Candidate Key

$(\texttt{student\_id})^+ = \{\texttt{student\_id}\}$

Apply FD1: $(\texttt{student\_id})^+ = \{\texttt{student\_id, full\_name, dob, gender}\}$

Since `student_id` uniquely identifies all atomic attributes, it is the **Candidate Key**.

7.2.3 Violation Identified

The attribute `phone_numbers` contains multiple values per student. 1NF strictly disallows multi-valued attributes in a single tuple.

7.2.4 Decomposition – Before 1NF (Violation)

```
STUDENT_UNNORMALIZED
+-----------+-----------+-----+--------+--------------+
|student_id |full_name  | dob |gender  |phone_numbers |
+-----------+-----------+-----+--------+--------------+
    (PK)
     |           ^         ^      ^            |
    FD1  --------+---------+------+   (VIOLATION: Multivalued)
```

7.2.5 Decomposition – After 1NF

```
STUDENT
+-----------+-----------+-----+--------+
|student_id |full_name  | dob |gender  |
+-----------+-----------+-----+--------+
    (PK)
     |           ^         ^      ^
   FD1a ---------+---------+------+

STUDENT_PHONE
+-----------+--------------+
|student_id |phone_number  |
+-----------+--------------+
  (PK,FK)         (PK)
     ^             ^
     +--Comp PK---+
```

Now every attribute contains single atomic values and repeating groups are removed. The relation satisfies **First Normal Form (1NF)**.

7.3 Step 2: Second Normal Form (2NF)

Rule: A relation must be in 1NF, and every non-prime attribute must be fully functionally dependent on the *entire* primary key (no partial dependencies).

7.3.1 Relation and Functional Dependencies

Relation: APP_EXAM(application_id, exam_id, exam_name, exam_date, total_marks)

Functional Dependencies:

$$\text{FD1:}\quad (\texttt{application\_id, exam\_id}) \rightarrow \texttt{exam\_name,}\ \ \texttt{exam\_date,}\ \ \texttt{total\_marks}$$
$$\text{FD2:}\quad \texttt{exam\_id} \rightarrow \texttt{exam\_name,}\ \ \texttt{exam\_date,}\ \ \texttt{total\_marks}$$

7.3.2 Closure to Find Candidate Key

$(\texttt{application\_id, exam\_id})^+ = \{\texttt{application\_id, exam\_id}\}$

Apply FD2: $= \{\texttt{application\_id, exam\_id, exam\_name, exam\_date, total\_marks}\}$

Candidate Key: (`application_id`, `exam_id`)

7.3.3 Violation Identified

FD2 causes the violation. `exam_name`, `exam_date`, and `total_marks` depend only on `exam_id`, which is a part of the composite candidate key. This is a **partial dependency**.

7.3.4 Decomposition – Before 2NF (Violation)

```
APP_EXAM
+---------------+---------+-----------+-----------+------------+
|application_id |exam_id  |exam_name  |exam_date  |total_marks |
+---------------+---------+-----------+-----------+------------+
       (PK)         (PK)
        |            |          ^           ^           ^
      FD1 +----------+----------+-----------+------------+
                     |          ^           ^           ^
                    FD2(VIOLATION: Partial Dependency)---+
```

7.3.5 Decomposition – After 2NF

```
APPLICATION_RECORD
+---------------+---------+
|application_id |exam_id  |
+---------------+---------+
     (PK,FK)       (PK,FK)
        ^             ^
        +--Comp PK---+

EXAM
+---------+-----------+-----------+------------+
|exam_id  |exam_name  |exam_date  |total_marks |
+---------+-----------+-----------+------------+
   (PK)
    |           ^           ^            ^
  FD2a ---------+-----------+------------+
```

Each relation now contains attributes that depend fully on its primary key. All relations satisfy **Second Normal Form (2NF)**.

7.4 Step 3: Third Normal Form (3NF)

Rule: A relation must be in 2NF, and no non-prime attribute can be transitively dependent on the primary key.

7.4.1 Relation and Functional Dependencies

Relation: APP_CENTER(application_id, center_id, center_name, city, state)

Functional Dependencies:

FD1: $\texttt{application\_id} \rightarrow \texttt{center\_id}$

FD2: $\texttt{center\_id} \rightarrow \texttt{center\_name, city, state}$

7.4.2 Closure to Find Candidate Key

$(\texttt{application\_id})^+ = \{\texttt{application\_id}\}$

Apply FD1: $= \{\texttt{application\_id, center\_id}\}$

Apply FD2: $= \{$`application_id, center_id, center_name, city, state`$\}$

Candidate Key: `application_id`

7.4.3 Violation Identified

FD2 causes the violation. `application_id` determines `center_id`, which in turn determines `center_name` and `city`. A non-key attribute determines other non-key attributes – a **transitive dependency**.

7.4.4 Decomposition – Before 3NF (Violation)

```
APP_CENTER
+----------------+-----------+-------------+------+-------+
|application_id  |center_id  |center_name  |city  |state  |
+----------------+-----------+-------------+------+-------+
       (PK)
        |              ^           ^          ^       ^
       FD1 ------------+           |          |       |
                       |           ^          ^       ^
                       FD2(VIOLATION: Transitive Dependency)
```

7.4.5 Decomposition – After 3NF

```
APPLICATION_CENTER
+----------------+-----------+
|application_id  |center_id  |
+----------------+-----------+
       (PK)          (FK)
        |              ^
       FD1a -----------+

CENTER
+-----------+-------------+------+-------+
|center_id  |center_name  |city  |state  |
+-----------+-------------+------+-------+
    (PK)
     |             ^          ^       ^
   FD2a -----------+----------+-------+
```

No non-key attribute depends on another non-key attribute. All relations satisfy **Third Normal Form (3NF)**.

7.5 Step 4: Boyce-Codd Normal Form (BCNF)

Rule: For every functional dependency $X \to Y$, X must be a superkey.

7.5.1 Relation and Functional Dependencies

Relation: EVALUATION(application_id, subject, evaluator_id)

Functional Dependencies:

$$\text{FD1:} \quad (\texttt{application\_id}, \texttt{subject}) \to \texttt{evaluator\_id}$$
$$\text{FD2:} \quad \texttt{evaluator\_id} \to \texttt{subject}$$

(An evaluator grades only one specific subject.)

7.5.2 Closure to Find Candidate Keys

Test 1: $($`application_id, subject`$)^+$
Start: $\{$`application_id, subject`$\}$
Apply FD1: $\{$`application_id, subject, evaluator_id`$\} \Rightarrow$ **Candidate Key 1**

Test 2: (application_id, evaluator_id)$^+$
Start: {application_id, evaluator_id}
Apply FD2: {application_id, evaluator_id, subject} $\Rightarrow$ **Candidate Key 2**

7.5.3 Violation Identified

FD2 causes the violation. In evaluator_id $\rightarrow$ subject, the determinant evaluator_id is not a superkey (it is only part of Candidate Key 2). BCNF requires all determinants to be superkeys.

7.5.4 Decomposition – Before BCNF (Violation)

```
EVALUATION
+----------------+---------+--------------+
|application_id  |subject  |evaluator_id  |
+----------------+---------+--------------+
       (PK)          (PK)
        |             |          ^
       FD1 -----------+----------+ (Valid: Determinant is Superkey)
                            ^     |
                            +--FD2+ (VIOLATION: NOT a Superkey)
```

7.5.5 Decomposition – After BCNF

```
EVALUATOR_SUBJECT
+--------------+---------+
|evaluator_id  |subject  |
+--------------+---------+
     (PK)
      |             ^
    FD2a -----------+
 (evaluator_id is now sole PK => valid Superkey)

APPLICATION_EVALUATOR
+----------------+--------------+
|application_id  |evaluator_id  |
+----------------+--------------+
     (PK,FK)         (PK,FK)
        ^              ^
        +---Comp PK----+
```

All determinants are now superkeys. All relations satisfy **Boyce-Codd Normal Form (BCNF)**.

7.6 Lossless Join Property

A decomposition is lossless if the original relation can be reconstructed by joining the decomposed relations.

For example:

$$COURSE \cap DISCIPLINE = \{\texttt{discipline\_id}\}$$

Since discipline_id is the primary key of DISCIPLINE, the decomposition is lossless.

Similarly:

$$\begin{aligned} ENROLLMENT \cap LEARNER &= \{\texttt{learner\_id}\} \\ ENROLLMENT \cap COURSE &= \{\texttt{course\_id}\} \\ RESULT \cap ENROLLMENT &= \{\texttt{enrollment\_id}\} \\ PAYMENT \cap EXAM\_REGISTRATION &= \{\texttt{exam\_reg\_id}\} \end{aligned}$$

Since these attributes are primary keys in their respective relations, all decompositions satisfy the lossless join property.

7.7 Dependency Preservation

Dependency preservation ensures that all functional dependencies can be enforced without joining relations.

The dependencies are preserved in their respective relations:

- `learner_id` → *learner attributes* (LEARNER)
- `course_id` → *course attributes* (COURSE)
- `discipline_id` → *discipline attributes* (DISCIPLINE)
- (`learner_id`, `course_id`) → *enrollment attributes* (ENROLLMENT)
- (`enrollment_id`, `assignment_id`) → *submission attributes* (ASSIGNMENT_SUBMISSION)
- `enrollment_id` → *exam registration attributes* (EXAM_REGISTRATION)
- `exam_reg_id` → *payment attributes* (PAYMENT)
- `enrollment_id` → *result attributes* (RESULT)
- `result_id` → *certificate attributes* (CERTIFICATE)
- `grievance_id` → *grievance attributes* (GRIEVANCE)

Thus, dependency preservation is maintained.

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7.8 Final Normalized Schema

The final normalized database schema for NPTEL, including both Primary/Foreign Key designations and Functional Dependency (determines) arrows, is presented below in five groups. The schema satisfies:

$$\text{1NF} \rightarrow \text{2NF} \rightarrow \text{3NF} \rightarrow \text{BCNF}$$

with lossless join and dependency preservation.

7.8.1 Group 1: User & Institution Profiles

```
LEARNER
+------------+-----------+-------+-----+--------+-----------------+------------------+--------------------+---------------+-------------------+
| learner_id | full_name | email | dob | gender | education_level | institution_name | state_of_residence | password_hash | registration_date |
+------------+-----------+-------+-----+--------+-----------------+------------------+--------------------+---------------+-------------------+
     (PK)
      |            ^         ^      ^      ^             ^                 ^                   ^                  ^                 ^
FD1   +------------+---------+------+------+-------------+-----------------+-------------------+------------------+-----------------+

LEARNER_PHONE                                   INSTITUTION
+------------+--------------+                   +----------------+------------------+------+---------+
| learner_id | phone_number |                   | institution_id | institution_name | city | website |
+------------+--------------+                   +----------------+------------------+------+---------+
   (PK,FK)        (PK)                                (PK)
      ^            ^                                   |                  ^            ^        ^
      +--Comp PK---+                             FD2   +------------------+------------+--------+

INSTRUCTOR
+---------------+-----------+-------------+-------+-------+----------------+
| instructor_id | full_name | designation | email | phone | institution_id |
+---------------+-----------+-------------+-------+-------+----------------+
      (PK)                                                  (FK)
       |              ^            ^          ^       ^           ^
FD3    +--------------+------------+----------+-------+-----------+

EVALUATOR
+--------------+-----------+---------------+-------+-------+
| evaluator_id | full_name | qualification | email | phone |
+--------------+-----------+---------------+-------+-------+
      (PK)
       |             ^            ^           ^       ^
FD4    +-------------+------------+-----------+-------+
```

7.8.2 Group 2: Academic Catalog

```
DISCIPLINE
+---------------+-----------------+-------------+
| discipline_id | discipline_name | description |
+---------------+-----------------+-------------+
      (PK)
       |                ^               ^
FD5    +----------------+---------------+

COURSE
+-----------+-------------+---------------+----------------+----------+------+-------------------+------------+-----------+--------+
| course_id | course_name | discipline_id | duration_weeks | semester | year | total_assignments | best_of_n | exam_date | status |
+-----------+-------------+---------------+----------------+----------+------+-------------------+------------+-----------+--------+
    (PK)                        (FK)
     |            ^              ^              ^            ^        ^           ^               ^           ^          ^
FD6  +------------+--------------+--------------+------------+--------+-----------+---------------+-----------+----------+

COURSE_INSTRUCTOR                             WEEKLY_ASSIGNMENT
+-----------+---------------+------+          +---------------+-----------+-------------+--------------+----------+-----------+
| course_id | instructor_id | role |          | assignment_id | course_id | week_number | release_date | due_date | max_marks |
+-----------+---------------+------+          +---------------+-----------+-------------+--------------+----------+-----------+
  (PK,FK)       (PK,FK)                             (PK)          (FK)
     ^             ^          ^                      |             ^           ^            ^             ^          ^
     +---Comp PK----+         |                FD8   +-------------+-----------+------------+-------------+----------+
FD7  +------------------------+
```

7.8.3 Group 3: Enrollment & Operations

```
ENROLLMENT
+---------------+------------+-----------+-----------------+------------------+--------+
| enrollment_id | learner_id | course_id | enrollment_date | assignment_score | status |
+---------------+------------+-----------+-----------------+------------------+--------+
      (PK)           (FK)        (FK)
       |              ^           ^              ^                 ^             ^
FD9    +--------------+-----------+--------------+-----------------+-------------+

ASSIGNMENT_SUBMISSION
+---------------+---------------+---------------+--------------+----------------+
| submission_id | enrollment_id | assignment_id | submitted_at | marks_obtained |
+---------------+---------------+---------------+--------------+----------------+
      (PK)            (FK)            (FK)
       |               ^               ^               ^               ^
FD10   +---------------+---------------+---------------+---------------+

EXAM_REGISTRATION
+-------------+---------------+----------+----------------+--------+
| exam_reg_id | enrollment_id | reg_date | preferred_city | status |
+-------------+---------------+----------+----------------+--------+
     (PK)           (FK)
      |              ^             ^             ^            ^
FD11  +--------------+-------------+-------------+------------+

PAYMENT
+------------+-------------+--------+--------------+----------------+----------------+--------------+
| payment_id | exam_reg_id | amount | payment_date | payment_status | transaction_id | payment_mode |
+------------+-------------+--------+--------------+----------------+----------------+--------------+
     (PK)        (FK)
      |            ^           ^           ^              ^                ^               ^
FD12  +------------+-----------+-----------+--------------+----------------+---------------+

GRIEVANCE
+--------------+---------------+----------------+----------------+----------------+--------+---------------+
| grievance_id | enrollment_id | grievance_type | grievance_text | grievance_date | status | resolved_date |
+--------------+---------------+----------------+----------------+----------------+--------+---------------+
      (PK)           (FK)
       |               ^                ^                ^                ^            ^           ^
FD13   +---------------+----------------+----------------+----------------+------------+-----------+
```

7.8.4 Group 4: Infrastructure & Allocation

```
TEST_CENTRE
+-----------+-------------+------+-------+----------+----------+
| centre_id | centre_name | city | state | capacity | operator |
+-----------+-------------+------+-------+----------+----------+
    (PK)
     |             ^          ^      ^        ^          ^
FD14 +-------------+----------+------+--------+----------+

HALL                                          HALL_ALLOCATION
+---------+-----------+-------------+----------+   +---------------+-------------+-----------+---------+-------------+
| hall_id | centre_id | hall_number | capacity |   | allocation_id | exam_reg_id | centre_id | hall_id | seat_number |
+---------+-----------+-------------+----------+   +---------------+-------------+-----------+---------+-------------+
   (PK)       (FK)                                        (PK)           (FK)         (FK)      (FK)
    |          ^             ^            ^                |              ^             ^         ^           ^
FD15+----------+-------------+------------+           FD16 +--------------+-------------+---------+-----------+
```

7.8.5 Group 5: Evaluation & Results

```
ANSWER_SHEET
+----------+-------------+--------------+-----------------+----------------+-------------------+
| sheet_id | exam_reg_id | evaluator_id | submission_date | marks_obtained | evaluation_status |
+----------+-------------+--------------+-----------------+----------------+-------------------+
   (PK)         (FK)          (FK)
    |            ^              ^               ^                 ^                  ^
FD17 +-----------+--------------+---------------+-----------------+------------------+

RESULT
+-----------+---------------+------------------+------------+-------------+------------------+-------------+
| result_id | enrollment_id | assignment_score | exam_score | final_score | certificate_type | result_date |
+-----------+---------------+------------------+------------+-------------+------------------+-------------+
    (PK)          (FK)
     |              ^                ^               ^            ^               ^                ^
FD18 +--------------+----------------+---------------+------------+---------------+----------------+

CERTIFICATE
+----------------+-----------+------------+----------------+--------------+
| certificate_id | result_id | issue_date | certificate_no | download_url |
+----------------+-----------+------------+----------------+--------------+
       (PK)          (FK)
        |              ^           ^               ^               ^
FD19    +--------------+-----------+---------------+---------------+
```

7.9 Reduction of Redundancy and Ensuring Data Integrity

One of the primary objectives of normalization in the National Examination Management System (NPTEL) is to reduce data redundancy and ensure data integrity. Redundancy refers to unnecessary duplication of data within database relations. Excessive redundancy leads to inconsistency, increased storage usage, and data anomalies.

7.9.1 Redundancy in Unnormalized Design

Consider an unnormalized relation:

```
LEARNER_COURSE(learner_id, full_name,
               course_id, course_name, discipline_name,
               centre_id, centre_name, city)
```

In this structure:

- Learner details are repeated for every course enrollment.
- Course details are repeated for every learner.
- Centre details are duplicated across multiple rows.

This results in high redundancy.

7.9.2 Update Anomalies

Redundancy causes the following anomalies:

1. **Update Anomaly:** If a course's exam date changes, it must be updated in every row containing that course. Failure to update all occurrences leads to data inconsistency.

   Example: If `exam_date` of a course is updated in one row but not in others, inconsistent data appears.

2. **Insert Anomaly:** A new Test Centre cannot be inserted unless at least one learner is assigned to it. This prevents independent insertion of entity information.

3. **Delete Anomaly:** If the last learner record associated with a Test Centre is deleted, the centre's information may also be lost. This leads to unintended data loss.

7.9.3 Elimination of Redundancy through Normalization

After normalization, relations are decomposed as:

```
LEARNER(learner_id, full_name, ...)    -- stored exactly once
COURSE(course_id, course_name, ...)    -- stored exactly once
DISCIPLINE(discipline_id, ...)         -- stored exactly once
TEST_CENTRE(centre_id, ...)            -- stored exactly once
ENROLLMENT(enrollment_id, learner_id, course_id, ...)
```

Now:

- Learner details are stored only once in LEARNER.
- Course details are stored only once in COURSE.

- Centre details are stored only once in TEST_CENTRE.
- Discipline details are stored only once in DISCIPLINE.

Thus, redundancy is minimized.

7.9.4 Ensuring Data Integrity

Normalization ensures integrity in the following ways:

- **Entity Integrity:** Each relation has a primary key that uniquely identifies tuples.
- **Referential Integrity:** Foreign keys enforce valid references between relations.
- **Domain Integrity:** Attributes follow defined data types and constraints. Certificate types are restricted to defined NPTEL categories.
- **Business Rule Integrity:** UNIQUE constraint on (learner_id, course_id) in ENROLLMENT enforces the one-enrollment-per-course rule.

For example:

- `exam_reg_id` in PAYMENT references EXAM_REGISTRATION.
- `enrollment_id` in EXAM_REGISTRATION references ENROLLMENT.
- `course_id` in ENROLLMENT references COURSE.
- `discipline_id` in COURSE references DISCIPLINE.

These constraints prevent invalid or inconsistent data entries.

7.9.5 Impact on NPTEL Performance and Reliability

Since NPTEL handles millions of learner registrations and concurrent transactions:

- Reduced redundancy lowers storage overhead across millions of enrollment records.
- Smaller, focused relations improve query efficiency for operations such as score computation and certificate generation.
- Foreign key constraints prevent invalid registrations (e.g., registering for a proctored exam without an active enrollment).
- Normalized assignment submission records ensure accurate best-of-N score computation without duplication.
- Transaction consistency is maintained under concurrent operations during assignment deadlines and exam registration windows.

Therefore, normalization significantly enhances both reliability and performance of the system at national scale.

7.10 Normalization Schema Diagram

The following diagram illustrates the complete normalized schema of the National Examination Management System (NPTEL), showing all decomposed relations after applying 1NF, 2NF, 3NF, and BCNF.

[Normalized Schema Diagram – National Examination Management System (NPTEL)]

7.11 Conclusion

The National Examination Management System (NPTEL) has been systematically designed and analyzed to support the complete lifecycle of India's largest online higher education programme – from free course enrollment and weekly assignment submission through proctored certification exam management to certificate issuance and grievance resolution.

In Assignment–1, the NPTEL system requirements were carefully studied and translated into entities, attributes, and relationships specific to the NPTEL framework, including disciplines, courses, instructors, institutions, weekly assignments, exam registrations, Test Centres, and certificate categories. A high-level conceptual model was developed using an ER diagram representing the entire database structure. All constraints, including primary keys, foreign keys, participation constraints, and relationship cardinalities, were clearly identified. The ER model was then successfully converted into a relational schema using standard ER-to-Relational mapping rules.

In Assignment–2, functional dependencies were identified for each relation. Candidate keys and primary keys were determined based on these dependencies. The relations were normalized step-by-step from Unnormalized Form (UNF) to First Normal Form (1NF), Second Normal Form (2NF), Third Normal Form (3NF), and further verified for Boyce-Codd Normal Form (BCNF).

During normalization:

- Data redundancy was eliminated.
- Partial and transitive dependencies were removed.
- Insertion, deletion, and update anomalies were avoided.
- Lossless join property was maintained.
- Dependency preservation was ensured.

The final normalized schema guarantees:

- Strong entity integrity through primary keys across all 19 relations.
- Referential integrity through carefully designed foreign key constraints.
- Reduced storage overhead due to minimal redundancy across millions of learner and enrollment records.
- Accurate score computation (25% assignment + 75% proctored exam) through clean, normalized result records.
- Improved consistency during concurrent enrollment, assignment submission, and payment transactions.
- High reliability and performance for large datasets befitting a national MOOC platform.

Overall, the NPTEL database design satisfies Third Normal Form (3NF) and fully conforms to Boyce-Codd Normal Form (BCNF) across all relations. The schema is scalable, secure, and fully suitable for real-world implementation supporting millions of learners across India in every academic semester.

Thus, the system design effectively achieves the objectives of data integrity, concurrency control, performance optimization, and structured information management.